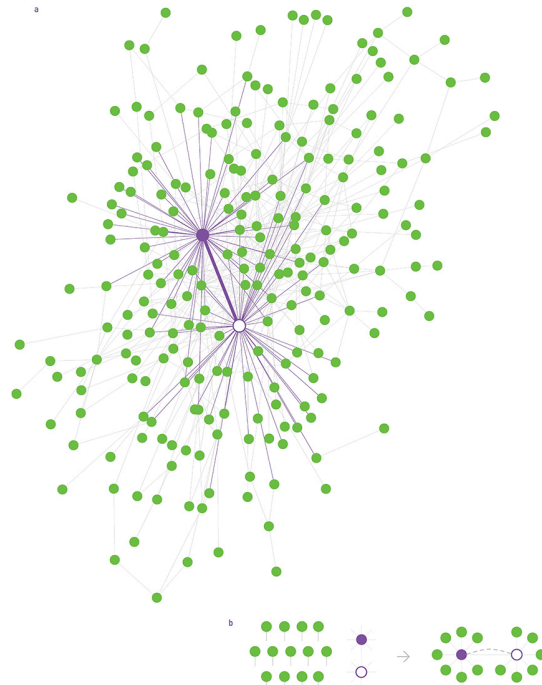


L4: The Average Degree of the Nearest Neighbor at a Power-law Network



Structural Disassortativity, Figure 7.7 from networksciencebook.com by Albert-László Barabási

Recall the notion of **“average neighbor degree”** from Lesson-3 – which is the same with the expected degree of a node that is connected to a randomly sampled edge stub.

Under the assumption of a **“neutral network”** (i.e., *no correlation between the degrees of two connected nodes*), we derived that the average neighbor degree is:

$$\bar{k}_n = \bar{k} + \frac{\sigma_k^2}{\bar{k}}$$

This shows one more interesting property of power-law networks: as we previously discussed, these networks can have very large degree variability in practice (i.e., σ_k is much higher than $\bar{k}$). Consequently, the average neighbor degree can be much higher than the network’s average degree – intensifying the impact of the friendship paradox.

The visualization at this page shows a power-law network with 300 nodes, 450 edges, and exponent 2.2, generated by the configuration model. We highlight the two highest-degree hubs. Note that many nodes are connected to those hubs, increasing the average neighbor degree of those nodes.